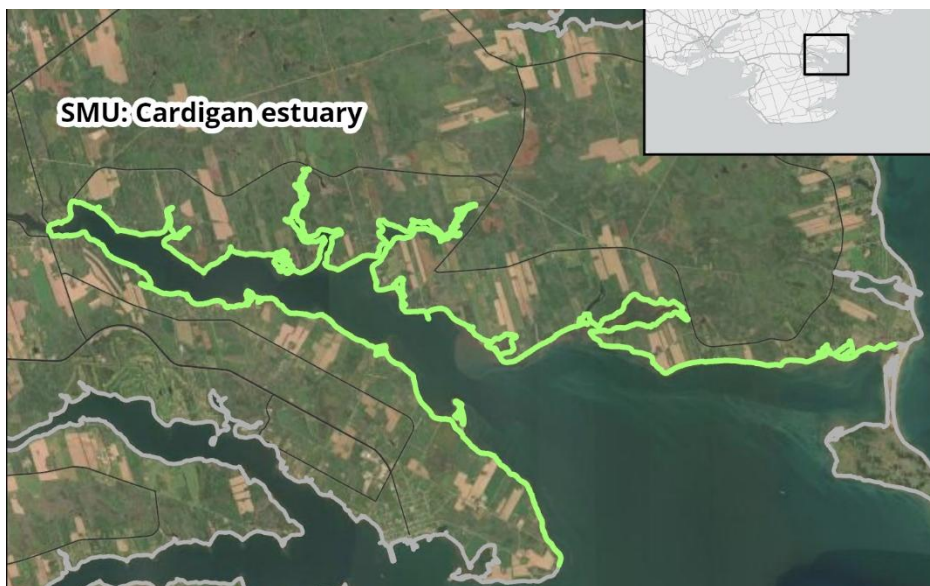


Cardigan Estuary

Shoreline length: 70 km



Shore Types

Cardigan Estuary is a tidal estuary with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
50	11	5	3	29	0	0	2

Governance, Infrastructure, and Economy

The Cardigan Estuary falls within the Town of Three Rivers.



Cardigan waterfront

The area includes coastal infrastructure including roads and facilities: the Cardigan River Lighthouse; Martell, Prestige, and Earl Power Roads (Georgetown, at the beach); Morrisons Beach Road (Georgetown); Cardigan Marina; Owens Wharf Road (Cardigan); Wharf Road (Route 321 bridge at Cardigan River); Chapel Road; Shore Road to Alleys Mill Road (Route 4, Cardigan); the Water Street bridge (77.2m, Cardigan); Seal River, Mitchell River, and School House Roads (Cardigan North); the Seal River Road bridge at Seal River, plus a parking lot; Launching Road (Route 311, DeGros Marsh); Farquarson Lane (DeGros Marsh); the Wharf Road wharf (293.5m); Mosquito Creek Road (DeGros Marsh, at the beach); Primrose Road (Route 311, across DeGros Marsh); and South DeGros Marsh Road (along the beach).

Coastal economic drivers include the PEI Coastal Sea Fish plant Stewart. Local and legacy land use constraints include several small-lot subdivisions serviced by unpaved roads, often perpendicular to the shore. Special consideration: the core area of Cardigan up to the bridge may warrant specific policy attention.

Natural Assets and Heritage

Parks and cultural facilities include the Kim Bujosevich Ballfield, Clipper Ballfield, Cardigan Park (playground), Cardigan Firehall (tennis and pickleball), the Cardigan River Lighthouse, Cardigan River Trails, the Cardigan Heritage Centre, and the Cardigan Library (Canada's smallest library). There are natural beaches and sandspits seaward of the estuary. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.



Bluff updrift of McPhee beach



Morrisons Beach

Planning and Policy Framework

Three Rivers has municipal planning authority. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Coastal Flood and Erosion Risk

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Cardigan Estuary				
Flood risk	10 [<1/km]	19 [<1/km]	31 [<1/km]	58 [1/km]
Erosion risk	N/A	31 [<1/km]	52 [1/km]	98 [1/km]

Sediment transport - The sediment transport rate averaged across the SMU is moderate and primarily directed inwards into the estuary. It feeds beaches and sandspits such as MacPhee Beach off Earl Power rd or Morrisons Beach.

Proposed SMP Strategies

Given the relatively low risk levels the initial strategy for Cardigan Estuary is No Active Intervention. In areas where infrastructure is at increasing risk, managed realignment should be considered in the mid- to long-term.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including wetlands and low plains with increasing flood hazard.		



LEGEND - Shoreline Management Unit - Net Longshore Sediment Transport - High Hazard Area - Flood Extent - Present Day (100-year Return) - Flood Extent - 1.2m SLR (100-year Return) - Property Boundary - Building > 90ft²	 0 325 650 1,300 Metres Coordinate System: NAD 83 CSRS PEI Double Stereographic NOTES:		COASTAL EROSION AND FLOOD HAZARD Shoreline Management Unit: Cardigan estuary	 <i>Île-du-Prince-Édouard</i>
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